

Virtual Global Landmark: An Augmented Reality Technique to Improve Spatial Navigation Learning

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ABSTRACT

Navigation is a multifaceted human ability involving complex cognitive functions. It allows the active exploration of unknown environments without becoming lost while enabling us to move efficiently across well-known spaces. However, the increasing reliance on navigation assistance systems reduces surrounding environment processing and decreases spatial knowledge acquisition and thus orienting ability. To prevent such a skill loss induced by current navigation support systems like Google Maps, we propose a novel landmark technique in augmented reality (AR): the virtual global landmark (VGL). This technique seeks to help navigation and promote spatial learning. We conducted a pilot study with five participants to compare the directional arrows with VGL. Our result suggests that the participants learned more about the environment while navigation using VGL than directional arrows without any significant mental workload increase. The results have a substantial impact on the future of our navigation system.

CCS CONCEPTS

• **Human-centered computing** → **Visualization toolkits; Displays and imagers**; • **Hardware** → **Emerging technologies**.

KEYWORDS

Spatial Navigation, Augmented Reality, Mental Workload, Navigation Landmark, Spatial Awareness

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1 INTRODUCTION

Spatial navigation is a process to explore an unknown environment and determine the route to a destination using spatial information. It has been well recognized that the capacities of spatial navigation and orientation are significant for all living-beings [1]. For human navigation support, popular navigation assistance systems such as Google Maps¹, Bing Maps², Apple Maps³, etc. have no doubt benefited us with efficient automatic orientation in everyday navigation. However, the increasing usage of these navigational supports can reduce surrounding environment processing and decrease spatial knowledge acquisition. Thus, the use of these systems has lead to a loss of navigational ability [4, 11, 16, 17] in their users. To overcome such effect, more cue sources about the environment are required during orientation to help users be aware of the surroundings to build up spatial ability. Improving the orientation and navigational skills directly impacts the neural connections due to the increase in the number of mental maps [10, 13]. Furthermore, previous works [2, 6, 15] have shown how landmarks are essential environmental cues that have a positive effect on navigation and spatial learning.

Given the benefit of a better navigation ability, considerable research comes across different techniques to improve spatial learning. In 2004, Richter and Klippel [19] presented an approach that adapted the route instruction based on the environment of the route to reduce the conceptualization and mental processing of route instructions. Later, Schmid et al. [20] developed a technique to generate route-aware maps similar to Richter et al. above, which help to understand the surrounding environment better. Recently, Li and colleagues [12] proposed a mobile map display to visualize distant landmarks for spatial orientation support, which helps people with a low sense of direction to be spatially oriented.

However, these techniques are helpful in learning, but due to presentation in two-dimensional screen, only useful for specific for people with specific reference frame proclivity [7]. Recent studies

¹<https://maps.google.com>

²<https://www.bing.com/maps>

³<https://www.apple.com/au/maps/>

have shown even richer proof towards the benefits of landmark cues that enhance spatial learning within virtual reality (VR) environments [3, 5, 14, 23]. These works, with the aid of recent significant development in portable, powerful, and wearable Augmented Reality (AR) technologies [21] opens up a new paradigm for orientation support. With AR technology, users can naturally interact with the pervasive computing landscape with digital information [18].

Current AR technologies allow displaying of digital information that can be used to place navigational icons such as arrows and paths in the real world. This seamless integration can improve the user's orientation and spatial navigation ability without creating confusion or discomfort. This could lead to a new method for spatial knowledge acquisition and neural processing. However, current AR-based navigation focuses on only overlaying landmarks that are visible to the user. This overlay has the intention of capturing the user's attention and not to aid in navigation.

The use of global landmarks as references is a well-known method to navigate unknown areas. For example, giving navigational directions to someone using as a reference the Sydney Opera House in Sydney, Australia, would be more helpful than explaining the same navigational directions with respect to other local landmarks. Motivated by this positive effect of global landmarks in spatial navigation ability, we present a novel landmark presentation schema, the Virtual Global Landmark (VGL), for an AR-based navigation system. Our contributions are as follows:

- (1) A novel virtual global landmark (VGL) technique allows seeing a global landmark even if its surroundings hinder it. This visibility update allows the user to use the VGL as a reference point with minimal interference to their natural navigation skill.
- (2) VGL can contribute to an improvement in spatial learning of users, which is shown to be essential to prevent long-term neurological issues such as memory loss [13].
- (3) A prototype of an AR-based navigation system that uses VGL as a navigation technique.

2 MATERIALS AND METHODS

2.1 Participant

The experiment was performed on five participants, all males. The mean age of the participants was 29 years, with a range of 26-33 years. Before participating in the study, each participant was given a full explanation of the experimental procedure, and each provided informed consent. Ethics approval was issued by the Human Research Ethics Committee of the University of Technology in Sydney, Australia. The experiment was conducted in a real-world environment. None of the participants had a history of any psychological disorders, which could have affected the experiment results.

2.2 AR setup

The AR application was built to run on a HoloLens 1 (Microsoft Corp., USA) device. This device comes with a field of view of 30°X17.5°. The participant's head direction was computed with an

inbuilt inertial measurement unit (IMU) and environment understanding cameras within the HoloLens. The AR application was built using the Unity 3D game engine ⁴.

Because the HoloLens system lacks a global positioning system (GPS), a mobile phone was used to provide the user location. A phone running the Android operating system (Samsung Galaxy A31) was used for this purpose. We developed a custom app in charge of reporting the user location to the AR system every time the phone's GPS location changes. To achieve this, the location updates were sent to a Google Firebase ⁵ database, which then sent the information to the HoloLens device.

2.3 Virtual Global Landmark

VGL is a directional aid to represent global landmarks virtually. In this experiment, the Sydney Opera House and the University of Technology, Sydney's main Tower Building, were used as global landmarks. The Opera House and Tower Building's virtual overlay image was presented in their actual angle of sight and location. The VGL will become visible to a participant if an actual landmark is not visible, which has been decided based on distance in the presented work. See Figure 1.

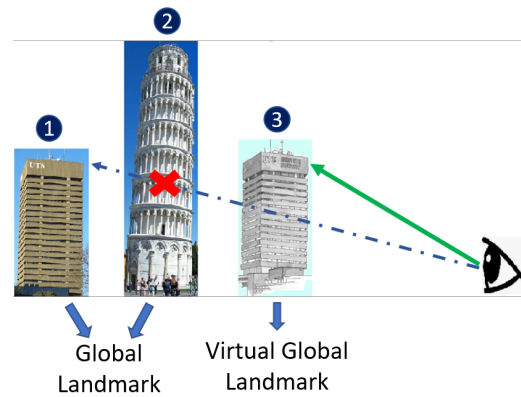


Figure 1: A VGL explanation : (1) an actual global landmark, (2) another landmark is hiding the first landmark, and (3) VGL representation of the first landmark in the sight of the landmark's direction.

2.4 VGL development

To display the VGL to the participant, the direction vector $\vec{vgl}$ was calculated between the participant and each of the real landmarks. The geolocation API Mapbox ⁶ was used for all the geographical calculations. We placed the VGL at the distance of one meter from the participant in the direction of $\vec{vgl}$. The VGL was always placed at the height of the participant's sight.

The VGL intends to give the participant a sense of seeing a floating hologram positioned in the real landmark direction. For that reason, a simple 2D drawing of the contour of each of the

⁴<https://unity.com/>

⁵<https://firebase.google.com/>

⁶<https://www.mapbox.com/>

buildings was used for each VGL. They were colored red to catch the participant's attention and set an 80% transparency to give it a hologram feel. Because the VGL was a 2D drawing, we made sure that the forward vector of the VGL was always pointing towards the participant. That way, the VGL was always visible no matter the viewing angle.

2.5 Experiment scenario

The experiment consists of four predefined navigation trials shown in Figure 2. Each trial consists of starting and destination points (marked as A and B in Figure 2). To successfully complete a trial, each participant needs to navigate from the starting point A to the destination point B. Participants navigated with the HoloLens device's help while the experimenter walked behind them within a distance of half a meter without any interruption. Once the participant was within a 10-meter distance of the destination, the HoloLens system will display a message that reads 'Destination Reached'. This message signaled the end of a trial. After this, the experimenter took the participant to the next start point for a new trial. Participants were instructed to walk naturally, avoiding any traffic and people during the walk, like in a normal navigational situation.

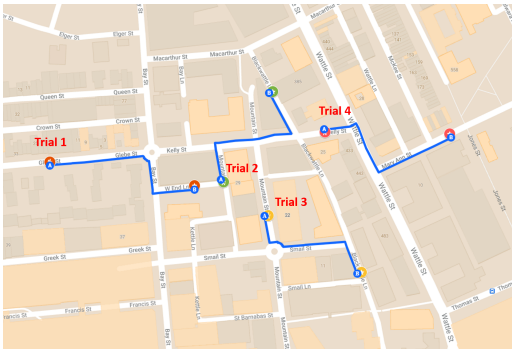


Figure 2: Navigation routes used for each trial for each participant. The map shown here shows the suburbs of Glebe and Ultimo in Sydney, NSW, Australia. The starting point of routes is represented by A and the destination by B.

Our navigation system contains two navigational aids: directional arrow and VGL. In each trial, the participant navigated using only one of those. The methodology was assigned randomly, making it two trials of the 'directional arrow' and two trials of 'VGL'. Besides these techniques, the user was always shown a map that showed the start point, the destination point, a suggested route to follow, and the user's current location. Figure 3 shows the AR screen setup from the user's point of view. All participants were instructed to use this map as a reference if necessary. For each navigation trial, the user walked a distance of around 350 meters. Each trial took about seven minutes on average.

2.6 Questionnaires

Before starting any navigation trial, we asked each participant to answer two 7-point Likert scale-based questionnaires. The first



Figure 3: Experiment design: (left) a trial consists of VGL for navigation with a map in the middle; (right) a trial consists of directional arrows for navigation with a map in the middle; This picture is for representational purpose only, to illustrate the differences between VGL and directional arrows.

questionnaire was a NASA-TLX [8] workload questionnaire. This questionnaire contains 7-questions to understand the participant's level of mental workload before starting an experiment. The NASA-TLX questionnaire is followed by a 7-point Likert scale based on Santa Barbara Sense of Direction (SBSOD) Scale [9], which contains 15 questions to understand on participant's navigational skill. It is noted that although we have collected SBSOD but did not use it for further analysis.

The NASA-TLX was also asked at the end of each trial. We wanted to understand if the participants' workload level changed due to the different navigational aids (directional arrow or VGL). Also, between each trial, we asked each participant to draw the whole navigation route. We asked them to specify any landmarks that they consider important such as shops, peculiar houses, unique trees, among others. At the end of the experiment, we asked the participants to fill a questionnaire on preference between 'directional arrow' and 'VGL'. Overall, it took about an hour to complete the experiment, including all navigation trials, questionnaires, and map drawing.

2.7 Statistics

Due to the ordinal scale in NASA-TLX, map drawing, and the preference questionnaire, we have analyzed the data using non-parametric Wilcoxon signed-rank test [22] at the 10% significance level. We have applied a 10% significance level due to a very smaller sample size in this study.

2.8 Hypotheses

The objective of conducting this experiment is to evaluate the efficacy of VGL compared to traditional means of navigation (directional arrow). We believe that VGL can improve the users' spatial learning and awareness about the environment compared to directional arrows. Therefore, we formulated the following hypotheses:

- **H1:** Users may require higher cognitive effort to use VGL than directional arrows; therefore, there will be a higher mental workload in VGL trials than the directional arrow trials.
- **H2:** Participants will be more spatially aware during navigation with VGL compared to the directional arrows.

3 RESULTS AND DISCUSSION

The results from NASA-TLX, map drawing, and personal preferences are presented in Figure 4.

Mental workload during navigation: As shown in Figure 4 (A), overall mental workload slightly increased for both VGL and directional arrow trials. However, VGL trials invoked lower but statistically non-significant ($p > 0.05$) mental workload compared to directional arrow trials. This might be due to two actions that are required from the user when using the directional arrows. First, they need to locate the position of the directional arrow. Second, they need to identify where the arrow is pointing to. In the case of the VGL trials, the participants simply had to check the direction of the destination compared to the location of the VGL. Therefore, the mental workload was found to be lower than the directional arrow.

The mental workload for directional arrow was found to be significantly higher ($p < 0.05$) compared to the pre-experiment and trial questionnaire. We believe this happened because all our participants had no experience using the HoloLens device. We believe that getting used to the device introduced some additional mental workload to all participants. These results between VGL and directional arrow do not support our hypothesis H1, which says that VGL may require higher cognitive workload to process the environment and find the direction compared to the directional arrow. The detailed score for each participant is presented in Supplementary Table 1.

The figure 4 (B) shows the individual components contributing to mental workload. It can be seen that the participants required higher mental demand, performance, and effort during the directional arrow trials compared to VGL trials ($p > 0.05$). However, directional arrow trials induce statistically significantly higher mental demand ($p < 0.10$), performance ($p < 0.10$), and effort ($p < 0.05$) requirements compared to pre-trial conditions. VGL also induced higher mental demand, performance, and effort, but these elements were statistically non-significant ($p > 0.10$). Interestingly pre-trial temporal demand was higher than VGL and directional arrow trials, but the results were not statistically significant ($p > 0.10$). One reason why the difference was non-significant could be that there was no time pressure on the participant to complete the trial or to reach on time at a certain location.

Our results show how VGL trials required more physical demand ($p > 0.10$) and produced a higher level of frustration compared to the pre-trial ($p > 0.10$) and directional arrows ($p > 0.10$) conditions. Similarly, directional arrow trials were found to produce a significantly higher ($p < 0.10$) level of physical demand and frustration. We found this obvious because the task demand itself required lots of physical effort to find the destination. In some cases, this task created frustration in our participants because they could not judge the correct location correctly. Interestingly, physical demand and frustration were higher for VGL compared to the directional arrow. One probable reason could be the additional demand for the physical effort to relocate in respect to the VGL followed by finding the destination. Again, these results partially do not support hypothesis H1, suggesting higher cognitive workload for VGL than directional arrows. Also, see Supplementary Table 2 for a full score for each participant.

Spatial learning: Our results support hypothesis H2 for spatial awareness between VGL and directional arrow condition following

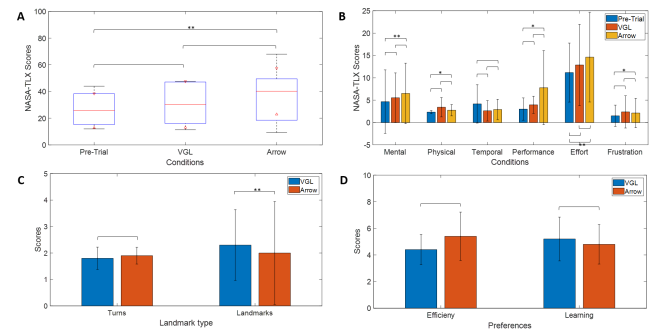


Figure 4: Questionnaire result from NASA-TLX, map-drawing, and participant preference toward 'directional arrow' or 'VGL': A) Overall NASA-TLX score representing mental workload for pre-trial, VGL, and directional arrow condition; B) Independent component contributed to overall mental workload using NASA-TLX for pre-trial, VGL, and directional arrow condition; C) total score for VGL and directional arrow trials based on a map drawing on a number of correct turns and number of correctly identified landmark; D) participant's preference toward VGL and directional arrow based on efficiency/easy to use and effectiveness toward learning about navigation environment (*: $p < .10$, **: $p < .05$, ***: $p < .01$).

mental workload results. As shown in the Figure 4 (C), it is evident that participants were able to remember significantly ($p < 0.05$) more number of landmarks for VGL trials compared to directional arrow trials. These results directly support our hypothesis H2, suggesting that VGL assisted navigation will encourage the participants to be more spatially aware compared to traditional navigation means such as directional arrows. The conventional navigation assistance aids such as directional arrows were beneficial for finding a destination correctly. However, users were discouraged from processing the information from the surrounding environment.

Interestingly, both VGL and directional arrow trials showed an almost similar ($p > 0.10$) number of turns recalled by the participant. One apparent reason is the length of the route, which almost always consists of two turns. Therefore, nearly every participant could draw them correctly during the map-drawing task at the end of the trials. The total number of turns and number of landmarks has been presented in Supplementary Table 3.

Preferences on learning and ease of use: After evaluating VGL and directional arrows' advantages and disadvantages using NASA-TLX and map-drawing, we also asked the participants to rate their preference for VGL and directional arrows. The user rated the techniques on a binary mark of efficiency/ease of use and spatial ability awareness. As shown in Figure 4 (D), in terms of efficiency/ease of use, participants always prefer a directional arrow over VGL ($p > 0.10$). Please also see Supplementary Table 4. Users found the directional arrow is easy to follow and helpful when reaching the destination. On the other hand, users found that VGL required additional effort, which is why participants preferred directional arrows over VGL. However, on the question of spatial learning, the participants chose VGL, which again aligned with hypothesis H2.

VGL indeed helps in spatial learning, which is also supported by the results from the map drawing. The participant rating has been presented in Supplementary Table 4.

Although this study showed that VGL is useful for improving navigation ability, we found this study has several limitations as follows:

- (1) The number of participants in the current study is only five, which is too small to get enough statistical power to solidify our claims. In the future, we plan to extend this study with a large number of participants to get enough power to support better conclusions.
- (2) We used a HoloLens version 1, which is not designed to work outdoors. Therefore, we found a significant problem when visualizing the map, the directional arrows, and the VGL. Consequently, we had to conduct all experiments on cloudy days for better visibility in HoloLens. In the future, we want to move from HoloLens to a better AR navigation device to avoid such issues.
- (3) We also faced other visualization issues such as jittering in maps, arrows, or VGL during walking, which is sometimes irritating for participants. In the future, we will make sure to avoid such issues using eye-tracking technologies similar to those in HoloLens 2.
- (4) The presented work highly relied on the participants' subjective rating, which could misinform the factual information. In the future, we plan to use physiological measures such as electroencephalogram (EEG), heart-rate variability (HRV), and/or pupil-size as measurements for mental-workload and learning ability to evaluate and assess VGL and directional arrows' performance objectively.

4 CONCLUSION

This work has developed a unique global landmark-represented AR technique known as VGL, which showed improvements in spatial learning. We have designed a real-world study that compares traditional directional arrow based navigational aids with a novel VGL technique. To do so, the participants were asked to navigate four pre-defined routes using either a directional arrow or VGL and then asked to answer NASA-TLX, draw the navigated route map, and easy to use learning preferences. The results show that VGL does not add much mental workload to the participant while promoting spatial learning compared to the directional arrows.

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